

Radar in the Dock: Were the Lakenheath–Bentwaters (1956) and RB-47 (1957) Tracks Genuine Solid-Object Returns, or Reducible to Technical Artifacts?

TL;DR

- **Neither case delivers a radar-derived, contemporaneously logged measurement of instantaneous acceleration (observable #2) or sustained anomalous speed without signatures (observable #3) that survives technical critique.** Every quantitative speed/acceleration figure is an operator's eyeball estimate off a PPI scope or a back-calculation by later analysts from sparse, memory-reconstructed plots — not an instrument-recorded kinematic measurement.
- **Lakenheath–Bentwaters has a genuine unexplained residue** (the Lakenheath "tail-chase" tracked concurrently on two ground radars of

different frequency/scan plus a Venom's airborne radar and pilot visual) that resists a clean single-artifact explanation, but it rests heavily on a 1968 memory letter and an AGC/MTI-retrofitted radar environment that skeptics plausibly argue could manufacture moving false targets.

- **The RB-47 case is the weaker of the two evidentially:** it has *no* surviving radar films, scope photos, or wire recordings — only a contemporaneous 1957 Wing intelligence teletype plus 1967 memories — and the most diagnostic "signature," the S-band ECM signal, matches a common ground radar (AN/CPS-6B) so closely that the skeptical (Klass/Printy) reduction is at least as strong as the anomalist (McDonald/Sparks) reading.

Key Findings

1. **All "speeds" are estimates, not measurements.** At Bentwaters, the initial GCA (AN/MPN-11A) track was logged by Airman 2nd Class John L. Vaccare Jr., who reported to T/Sgt L. Whenry a target covering roughly 40–50 miles in 30 seconds — i.e., 4,800 to 6,000 mph — while the operator's verbal estimate was "4,000 mph"; the ~10 p.m. track "covered 55 miles in just 16 seconds," over 12,000 mph. The "instantaneous" ~18,000 mph figure for URE No. 4 comes from a single PPI

sweep with a 180° antenna rotation. These are inferences from a coarse, intermittently-sampled scope, not tracked kinematics.

2. **The Condon Report (Case 2, Gordon David Thayer) genuinely could not explain the Lakenheath phase.** Its formal conclusion (Scientific Study of Unidentified Flying Objects, 1968): "although conventional or natural explanations certainly cannot be ruled out, the probability of such seems low in this case and the probability that at least one genuine UFO was involved appears to be fairly high." In Chapter 5 (p. 245) Thayer wrote: "In summary, this is the most puzzling and unusual case in the radar-visual files. The apparently rational, intelligent behavior of the UFO suggests a mechanical device of unknown origin as the most probable explanation of this sighting. However, in view of the inevitable fallibility of witnesses, more conventional explanations of this report cannot be entirely ruled out."
3. **Thayer's anti-AP argument is technically sound for the Lakenheath phase but weaker at Bentwaters.** He conceded anomalous propagation "may possibly have been a factor" in the slow Bentwaters cluster (URE No. 2), but argued the stop-go maneuvers tracked on two

different-frequency, different-scan-rate radars at Lakenheath were "almost inconceivable" as AP.

4. The strongest skeptical counter is not AP but the freshly-retrofitted MTI/AGC radars.

The Condon Report itself records (p. 245): "Philip Klass (private communication) believes that the Lakenheath RATCC radar was malfunctioning because of a faulty MTI unit; he feels that once the radar evidence has been explained, the rest can be accounted for by either confusion of witnesses or conventional causes." Critics note that newly-installed MTI in 1956 (as in the 1952 Washington National flap) could amplify weak AP/weather returns into "plane-like" blips.

5. The Lakenheath case substantially rests on a 1968 memory letter.

Watch Supervisor T/Sgt Forrest D. Perkins wrote to the Condon Committee (received February 1968) — nearly 12 years after the event. His account was later found broadly consistent with a contemporaneous classified teletype (the IR-1-56 / "BOI-485" intelligence documents), which is the case's main saving grace, but the most vivid details (the tail-chase, instantaneous accelerations) trace to memory.

6. For the RB-47, the diagnostic signal matched a CPS-6B ground radar almost exactly.

The 1957 Piwetz Wing Intelligence report (Capt. E.T. Piwetz,

55th SRW) recorded the signal detected near Meridian, Mississippi as: frequency 2,995–3,000 MHz; pulse width 2.0 μ s; PRF 600 pps; scan rate 4 rpm; vertical polarity. The AN/CPS-6B's six beams spanned 2,700–3,019 MHz with 1.0–2.0 μ s pulses and 2–15 rpm rotation – i.e., the "UFO" emitted exactly what a common S-band air-defense radar emits.

7. The core RB-47 anomaly is the "up-scope" motion and the claimed simultaneity across three channels. McClure's ALA-6/APR-9 monitor showed a ~3,000 MHz signal moving "up-scope" (implying an airborne, overtaking emitter) when the aircraft was in straight-and-level flight; later, signals reportedly blinked on/off simultaneously with ground radar (Duncanville "Utah") contact and visual sightings. The reconstruction is McDonald's, built from the 745th ACWRON (Duncanville) three-page TWX filed 1557Z, 17 July 1957, the Piwetz four-page summary, and Maj. Chase's 12-page Airborne Observer's Data Sheet (prepared 10 September 1957).

8. Klass and Printy reduce the RB-47 to parts: the up-scope event = a stuck relay in the ECM gear inverting the display of a ground CPS-6B (e.g., Keesler AFB); the first visual = a meteor (Brad Sparks agrees); the later radar/visual near Dallas = a landing airliner. Sparks rebuts that Keesler's

CPS-6B was a non-operating summer training radar and that the aircraft crossed the coast west of it; Printy counters that the training annex ran three 6-hour shifts with midnight maintenance retests and could have been radiating.

9. **A documentary contradiction undercuts the RB-47 simultaneity claim:** a redacted line in the Blue Book teletype reads "UTAH had negative contact with object," directly conflicting with the crew's account that Duncanville confirmed the target on its scopes.

10. **In both cases the visuals are confounded by prosaic stimuli:** the Perseid meteor shower had just passed maximum on the night of 13–14 August 1956 but was still within its high-rate window (the IMO places the shower's peak ~12–13 August), consistent with the Blue Book/Condon note that observers reported "an unusually large number of shooting stars"; a star/planet (Mars) and a lighthouse beacon (Orford Ness) were identified for some Bentwaters visuals; the RB-47's first light is conceded by both sides to likely be a meteor.

Details

CASE 1: Lakenheath–Bentwaters, 13–14 August 1956

The radar systems. Bentwaters operated an AN/MPN-11A GCA system (incorporating a CPN-4 surveillance radar, ~10 cm/S-band). Lakenheath's Radar Air Traffic Control Center (RATCC) used a CPS-5 (~23 cm/L-band, with on-site modifications increasing rotation rate and adding MTI), and the airfield GCA used a CPN-4. The Venom night fighter carried AI (airborne intercept) radar (variously cited as A-1 / APS-57). The crucial technical point: the Lakenheath CPS-5 and CPN-4 were recently retrofitted with MTI and AGC — the same recipe present in the 1952 Washington National radar flap, where (per skeptical analysts) AGC misadjustment caused UFOs to vanish from screens whenever a real aircraft entered the area.

The reported tracks. Four Bentwaters events (per Blue Book/McDonald):

- URE No. 1 (2130Z): picked up 25–30 mi ESE, constant 295° heading, lost 15–20 mi WNW. Operator said 4,000 mph; transit arithmetic gives 4,800–6,000 mph, with one sweep-to-sweep estimate as high as 9,000–10,800 mph.
- URE No. 2 (~2135–2155Z): a cluster of 12–15 slow returns (80–125 mph) merging into one return "several times larger than a B-36," hovering then moving off. **This is the one event Thayer**

conceded could be AP (the cluster moved with the prevailing SW wind at roughly half the apparent target speed, consistent with moving-layer partial reflection).

- URE No. 3 (2200Z): 30 mi E to 25 mi W in ~16 s → ~12,000 mph implied.
- URE No. 4 (2255Z): the event that triggered the call to Lakenheath; "disappeared 2 mi E of station, immediately appeared 3 mi W," implying (if the same sweep) ~18,000 mph; accompanied by ground-visual (control tower) and a C-47 pilot reporting a light passing beneath at ~4,000 ft. Thayer judged this *not* AP because the motion was nearly opposite the prevailing wind, but the Condon Report flagged the overhead "disappearance" as "suggestive of AP" (a ground radar's normal zenithal blind cone).

The **Lakenheath phase** (the evidential core): after switching to full MTI, controllers found a stationary echo 20–25 mi SW; it then accelerated to 400–600 mph with — in the operators' words — "no build-up to this speed — it was constant from the second it started to move until it stopped," made right-angle direction changes with 3–6 minute stationary episodes, and was confirmed on the GCA CPN-4 in the same location. A Venom was scrambled, obtained "gun-lock" ("the clearest target I have ever seen on radar"), the object reportedly slipped behind the

fighter into a tail-chase the pilot could not shake, then resumed a stationary aspect after he left. ("He — or it — got behind me and I did everything I could to get behind him and I couldn't. It's the damndest thing I've ever seen.")

Thayer's case against artifacts. (a) Meteors ruled out by the maneuvers and by observers distinguishing UFOs from the abundant Perseids. (b) AP ruled out for everything except possibly URE No. 2, on wind-direction grounds and because AP "ghost" echoes that tail aircraft never stop and go stationary. (c) MTI malfunction insufficient because the same target appeared on a *different* radar type (GCA CPN-4) and the Venom's airborne radar. (d) The apparently-stationary-on-MTI detection is explained because "a vibrating or rapidly rotating target will show up on MTI radar even if it is not otherwise in motion."

The skeptical rebuttals.

- **Klass (UFOs Explained, 1974):** the RATCC radar had a faulty MTI unit; once the radar is explained, the rest is witness confusion plus Perseid meteors and misidentified stars/planets. He cited a Bendix field engineer who had observed apparently fast-moving echoes that turned out to be equipment malfunctions, and suggested MTI delay-line problems on the CPN-4 could generate spurious fast echoes.

- **The MTI/AGC argument:** newly-installed MTI on vacuum-tube/mercury-delay-line technology in 1956 was buggy; AGC could amplify weak AP/weather/ground returns into aircraft-strength blips; once crews entered "UFO mode" (actively hunting), ordinary spurious returns became "UFOs."
- **Perseids:** the shower was just past peak; observers explicitly noted unusually numerous shooting stars.
- **Conflation of separate events:** the modern Lakenheath Collaboration (David Clarke, Andy Roberts, Martin Shough, Jenny Randles, Paul Fuller) concluded there was "not a single clear-cut incident but a somewhat complicated situation involving different incidents at different places and times." [Martinshough](#) They also found that whether radar and ground-visual observations at Lakenheath were ever actually *concurrent* is doubtful — the file's "two radar sets and three ground observers report substantially same" may indicate similarity, not simultaneity.

McDonald's work. Atmospheric physicist James E. McDonald recovered the Blue Book file (which Condon initially lacked), interviewed radar operators and the Venom pilot, and published "UFOs over Lakenheath in 1956" (Flying Saucer Review, Vol. 16, No. 2, 1970, pp. 9–17). He argued the L-/S-band

frequencies were ~2 orders of magnitude too high for meteor-wake or auroral-ionization returns and emphasized concurrent multi-radar tracking.

The reliability problem. The vivid narrative rests on Perkins's February 1968 letter (~12 years post-event). Its main corroboration is the contemporaneous IR-1-56/BOI-485 teletype (sent ~31 August 1956), which Shough's group says "support[s] Perkins' detailed testimony in every substantial respect." [Martinshough](#) But Flt. Lt. Freddie Wimbledon (RAF Neatishead controller) – who corroborated the chase in 1978 – also reassigned key control roles to the RAF (claiming his team, not Lakenheath, vectored the Venoms), and Shough's reconstruction shows real ambiguity about radar/visual concurrency. The newer fieldwork *deepened* rather than dissolved the mystery: Shough found no low-level inversion/ducting to explain the fast tracks, but neither did it produce contemporaneous logs of the maneuvers.

MTI-and-AP physics (1956). MTI works by canceling returns with zero/low Doppler (stationary clutter) and passing those with Doppler shift. Genuine 1950s failure modes that can mimic moving targets: (i) "second-time-around" (multiple-trip) echoes from beyond the unambiguous range, which on a low-PRF set can place a distant or fast object at a false near range; (ii) blind-speed artifacts; (iii) AGC misadjustment amplifying weak AP/ground returns;

(iv) AP itself from elevated inversion layers producing partial reflections that appear at twice the layer's range/height and move with the wind. The decisive constraints cut both ways: AP false targets move *with* the prevailing wind (the Lakenheath/Bentwaters fast tracks largely did not), and a single radar's artifact should *not* appear at the same place on a second radar of different frequency and scan rate — which is precisely the residue Thayer leaned on.

Verdict on Case 1. The fast Bentwaters single-radar tracks (#1, #3, #4) are individually vulnerable to multiple-trip echoes/AP/MTI artifacts and lack robust independent confirmation. The Lakenheath tail-chase is the genuine residue: if the two-ground-radar-plus-airborne-radar-plus-pilot-visual concurrency is real, no single artifact explains it. But that concurrency is exactly what is *least* well documented contemporaneously. Observables #2 and #3 are *asserted* (the "no build-up" acceleration) but never measured by instrument — they are operator descriptions off a ~15-second-sweep scope.

CASE 2: RB-47H, 17 July 1957

The electronic systems. The RB-47H carried three ELINT/ECM monitors. McClure's #2 monitor was an ALA-6 direction-finding receiver (back-to-back belly antennas spinning at 150 or 300 rpm, frequency range 1,000–7,500 MHz) feeding an APR-9 receiver and

ALA-5 pulse analyzer. [Ufocasebook](#) The #1 monitor was not on the relevant frequency; the #3 monitor could not tune it. The ECM gear was *passive* — it detected emitted signals; it did not skin-paint targets.

The detected signal. Per the contemporaneous Piwetz report, near Meridian, Mississippi the signal was 2,995–3,000 MHz, 2.0 μ s pulse width, 600 pps PRF, 4 rpm scan, vertical polarity. McClure separately told McDonald the *first* (coastal) signal was near 2,800 MHz. Crucially, the AN/CPS-6B air-defense radar (and its FPS-10 derivative) transmitted six beams spanning 2,700–3,019 MHz at 1.0–2.0 μ s and 2–15 rpm — so the "UFO's" emission was indistinguishable from a standard S-band ground radar. (One internal inconsistency, flagged by analysts: the recorded 2.0 μ s paired with 600 pps does not match the CPS-6B's published pairing of 2.0 μ s with 300 pps [Forumactif](#) — a genuine unexplained mismatch.) Printy characterizes 1957 Texas/Oklahoma as an "S-band minefield" of CPS-6B/FPS-10 sites (Keesler annex, Duncanville, Ellington AFS, OK City AFS, Bartlesville AFS) plus weather and airport-surveillance radars on overlapping frequencies.

The three-channel simultaneity claim. McDonald's reconstruction emphasizes several instances where the luminous object, the #2-monitor signal, and the Duncanville ("Utah") ground-radar return blinked off

and on together, and where the navigator's airborne radar reportedly showed the same bearing. The contemporaneous 1957 report states the object stopped, "Utah reported they lost object from scopes at this time and ECM nr 2 also lost signal," [Blogger](#) with re-acquisition on a port turn. The 55th SRW Director of Intelligence wrote he had "no doubt the electronic D/F's coincided exactly with visual observations by aircraft cmdr numerous times, thus indicating positively the object being the signal source."

The "up-scope" anomaly. A ground radar's signal should drift "down-scope" as the aircraft flies forward past it. McClure saw a signal move *up-scope* in straight-and-level flight (he checked with the cockpit that they were not turning) — implying an airborne, overtaking emitter. This is the part both McDonald and Sparks consider hardest to explain prosaically.

Condon's treatment (Case 5, Roy Craig): classed "unidentified," but only because "evaluation of the experience must rest entirely on the recollection of crew members ten years after the event" — there were no surviving films/recordings, and Craig had the wrong date (September vs. July 1957), so he never located the file. McDonald later corrected the date and recovered the documents.

The skeptical resolution (Klass, refined by Printy).

- Up-scope event = a stuck/faulty relay in the ECM display inverting a ground CPS-6B's signal so it appeared up-scope on the starboard side instead of down-scope on the port side.
- First visual = a meteor (Sparks concurs: "This meteor fireball sighting is the only part of the RB-47 incident having a mundane explanation, in this case as a natural phenomenon").
- Later Dallas-area radar/visual = an airliner (Klass nominated American Airlines flight #966 on approach with landing lights).
- The persistent S-band signal = misattributed ground CPS-6B(s).

Sparks's rebuttal and Printy's counter. Sparks argues (a) the aircraft crossed the coast near Gulfport, west of Keesler, making an up-scope geometry impossible; (b) Keesler's CPS-6B was a training set not operating in mid-summer middle-of-night; (c) the simultaneous $\sim 30^\circ$ -separation measurement of UFO vs. Duncanville signals proves the UFO was a distinct, dominant emitter — Sparks calls it "the first conclusive scientific proof for the existence of UFOs." Printy counters that the recorded precise parameters likely describe the *Texas/Duncanville* detection (where McClure actually wrote things down), not the coastal pass; that the Keesler annex ran three 6-hour shifts with midnight-shift maintenance retests and had classes graduating year-round (he documented graduation dates in

January, February, March, June, July, September, and October 1957), so the CPS-6B "COULD have been operating between 3 and 4 AM" to be ready for the 0600 class. Printy's honest caveat: "We will never know if the CPS-6B was operating that morning without the annex operating logs." His bottom line: the case is not UFOlogy's "best evidence," the electronic signature is fully consistent with terrestrial S-band radar, Sparks's rebuttals do not hold up, and the case is "still unidentified" only in the trivial sense of incomplete documentation — Klass's overall conclusions "still stand" despite some errors. Notably, even Klass's airliner identification is contested: critics report AA #966 was involved in a near-miss in west Texas and was nowhere near Dallas at the relevant time.

The fatal documentary problem. Even granting the crew's sincerity, a redacted Blue Book teletype line reads "UTAH had negative contact with object" — directly contradicting the claim that Duncanville confirmed the target. If Utah did not actually have the object on its scopes, the central "ground-radar + ECM + visual" triple-coincidence collapses to a two-channel (ECM + visual) event, both of which have prosaic candidates (ground CPS-6B + meteor/airliner/star). The meaning of the line is itself contested (no radar contact vs. no CIRVIS report filed).

Is moving up-scope from a stationary emitter

physically plausible? Yes, partially: as the RB-47 maneuvered (turns; the port turn near Mineral Wells), the relative bearing to a *fixed* ground emitter changes, producing up-scope/down-scope motion that is the aircraft's own motion, not the source's. The genuinely anomalous claim is up-scope motion during *straight-and-level* flight. A stuck relay inverting the display is Klass's answer to that specific sub-event. What neither the relay nor the bearing-geometry fully explains is the *simultaneous* blink-out across ECM + ground radar + visual — *if* that simultaneity is real. And that is precisely what is undocumented contemporaneously beyond the crew's own report and is contradicted by the "negative contact" teletype line.

Verdict on Case 2. The signature that should have been most diagnostic (the S-band emission) is the strongest evidence *against* an exotic object, because it is exactly a CPS-6B signal. The case's evidential spine — three-channel simultaneity — is reconstructed from ~10-year-old memory and partly contradicted by the contemporaneous file. No radar-derived measurement of instantaneous acceleration or signature-free hypersonic speed exists; the "maneuvers" are visual/bearing impressions.

Recommendations

For a researcher seeking to establish observables #2 and #3 rigorously:

1. Treat neither case as proof of instantaneous acceleration or signature-free hypersonic flight.

In both, every kinematic figure is an estimate off a coarse PPI or a memory reconstruction — not an instrument-logged track. The benchmark that would change this: a recovered radar film, scope-photo sequence, or wire/tape recording with timestamped range-azimuth plots. None exists for either case. If such a primary record surfaced (e.g., in UK MoD/RAF Neatishead archives for Lakenheath, or SAC/55th SRW or ADC records for the RB-47), re-open immediately.

2. Prioritize the Lakenheath tail-chase over

everything else, because it is the only sub-event with claimed multi-sensor, different-physics concurrency (CPS-5 + CPN-4 + airborne AI radar + pilot visual). The decisive open question is *temporal concurrency*. Seek: the RAF Neatishead GCI logs/tapes, the Venom squadron records (Waterbeach), and any RATCC log — to establish whether the ground-radar track and the airborne radar/visual were truly simultaneous or sequentially inferred. If concurrency cannot be documented, the residue weakens to "one well-described but singly-sourced episode."

3. **For the RB-47, resolve the "UTAH negative contact" contradiction first.** Everything hinges on whether Duncanville actually tracked the object. Seek the original (un-redacted) 745th ACWRON/Duncanville TWX and any ADC CIRVIS report. If Utah had negative contact, the case reduces cleanly to ECM-detected ground radar(s) + meteor/airliner/star, and should be reclassified as probably explained.
4. **Commission a modern radar-engineering simulation** of (a) 1956 MTI/AGC second-time-around and AP behavior for the CPS-5/CPN-4 against the actual 13–14 Aug 1956 refractivity profile, and (b) the ALA-6/APR-9 response to a CPS-6B at the RB-47's geometry, to quantitatively test the artifact hypotheses rather than argue them verbally.
5. **Weight sources thus:** contemporaneous (1956 IR-1-56/BOI-485 teletype; 1957 Piwetz report and Duncanville TWX; the redacted Blue Book teletype) > near-contemporaneous official (Condon 1968–69) > memory reconstructions (Perkins 1968; crew interviews 1967; Wimbledon 1978) > secondary advocacy (McDonald, Klass, Sparks, Printy). Anomalist and skeptic both over-claim: Sparks's "first conclusive scientific proof" is unsupportable; Klass's tidy three-part solution

depends on contested premises (an operating CPS-6B, an on-time airliner).

Caveats

- **Sourcing asymmetry.** Primary documents (the Condon text, Thayer's AIAA article, McDonald's papers, the 1957 Piwetz report) are well attested. Klass's and Sparks's full arguments were read here largely through faithful secondary reproductions (Kevin Randle's detailed transcriptions, the Lakenheath Collaboration site, and a focused reading of Printy's SUNlite 4-1). The verbatim final wording of Printy's conclusion and Klass's UFOs Explained chapter were not read in the original; they are reported via faithful secondary quotation.
- **Speed figures are not independent.** Different secondary sources cite overlapping but inconsistent speed ranges (2,000–4,000; 4,000; 4,800–6,000; 9,000–10,800; 12,000; 18,000 mph) because they apply different transit-time and sweep-rate assumptions to the *same* sparse data. The very spread is evidence that these are estimates, not measurements; the Bentwaters MPN-11A sweep rate itself is uncertain (cited variously as ~2 s, 6.4 s, or 10.66 s).

- **The "negative contact" teletype** is partly obscured/redacted in the file; its exact meaning (no radar contact vs. no CIRVIS filed) is itself contested, and is flagged rather than resolved here.
- **Both cases are over-determined by prosaic stimuli** (Perseids, Mars, the Orford Ness lighthouse, a meteor, a possible airliner), which means even genuine sensor anomalies are hard to cleanly isolate from coincident mundane triggers.
- This assessment deliberately does not adjudicate the extraterrestrial hypothesis; "genuine UFO" in Thayer's own usage meant only a material, moving, unidentified object — and even that conclusion he hedged.